

VAHE GDLYAN

Machine Learning Engineer · Data Scientist

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Machine Learning Engineer who ships production systems, not notebooks: live multi-cloud inference pipelines, dual-stage RAG architectures, and A/B testing frameworks validated across 887K users — built solo or as lead architect while still completing high school. Technical author on production ML failure modes and system design.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, Git, Docker, REST APIs

ML / DL: PyTorch, YOLOv8, LightGBM, XGBoost, Scikit-learn, HuggingFace, FashionCLIP (ViT-B/16), U-2-Net, CNNs, Vision Transformers, Gradient Boosting, Contrastive Learning

Backend & Infra: FastAPI, Redis, Celery, Modal.com (serverless GPU), Supabase (PostgreSQL + pgvector), ChromaDB, Cloudflare R2, DuckDB, Flutter

Experimentation: A/B testing, Welch's T-test, Z-test, SRM detection, Simpson's Paradox, Gini coefficient, Platt Scaling, zero-inflated modelling, custom loss functions, OOD analysis

SELECTED PROJECTS

NucleAI — AI-Powered Nuclear Engineering Platform

May 2026

Lead Engineer · Python · FastAPI · ChromaDB · OpenAI API · JavaScript

Award: Winner, HackAtom Armenia 2026

- Engineered a FastAPI RAG pipeline grounded on 10 IAEA PDFs (ChromaDB, text-embedding-3-small, k=4 cosine retrieval), achieving zero hallucination on in-distribution nuclear safety queries.
- Built a deterministic grid simulation engine — an 8-branch narrative cascade and a 4-component, 100-point scoring model (output, emissions, stability, diversification) — with zero external dependencies.

Vestimate — Multi-Cloud AI Wardrobe Platform

March 2026 – April 2026

Lead Architect & Co-Founder · FastAPI · Flutter · Modal.com · Supabase · Redis · Celery

Award: 1st Place, Plekhanov University Startup Competition 2026

- Architected a distributed ML system end-to-end: async FastAPI gateway → Redis/Celery queue → Modal NVIDIA T4 GPU (U-2-Net + FashionCLIP, ~1.2s/garment) → pgvector IVFFlat store (<15ms queries) — running in production at ≤\$80/month.
- Built a dual-stage RAG assistant with a Python hallucination guard (95% semantic consistency) and a Redis response cache (<80ms, 98%+ hit rate), with graceful fallback when the LLM is unreachable.

AgriVision — CV Decision Support System (Global Wheat Detection)

April 2026

Lead Engineer · YOLOv8 · PyTorch · Hugging Face

- Designed a focal boundary loss with adaptive photometric weighting that eliminated 1,092 zero-IoU hallucinated detections — preventing 4–8% yield overestimation — while holding 0.944 mAP@50, lifting median prediction confidence +39.4%, and cutting training epochs 30%.
- Authored a 7-page mathematical diagnostic paper on out-of-distribution boundary analysis.

CleanFrame — Out-of-Core Data Intelligence Engine

June 2026

Author · Python · DuckDB · Pandas · NumPy

- Built a streaming, out-of-core pipeline on DuckDB that processes 100GB+ datasets on dual-core CPU hardware with zero OOM failures, removing the need for cloud infrastructure in large-scale data cleaning.
- Designed a 'Do No Harm' diagnostic layer — automated ML target-leakage detection, business-logic enforcement, and fuzzy string unification — turning ad-hoc cleaning into a rigorously audited, reproducible process.

EDUCATION & TRAINING

Picsart Academy

Feb 2025 – May 2026

Machine Learning & Data Science Trainee · Production ML curriculum

Abovyan Educational Complex

Expected Jun 2026

High School Diploma · Focus: Mathematics & Computer Science · 2023 Vice Champion of Armenia, Junior Boxing